

Airbnb New User Bookings

Data Mining and Machine Learning Project

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Agenda

- 1 Problem & Data
- 2 Exploratory Data Analysis
- 3 Preprocessing & Feature Engineering
- 4 Modeling & Evaluation
- 5 Hierarchical Approach & Imbalance Handling
- 6 Ensembles & Final Results
- 7 Conclusions

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- Kaggle competition: **Airbnb New User Bookings**
- Task: **12-class classification** (first booking destination)
- Majority outcomes: NDF (no booking) and US
- Challenge: heavy **class imbalance** + mixed feature types + session logs

Context and Goal — Explanation

What the task involves

Each new Airbnb user must be assigned one of **12 destination labels**: NDF (no booking yet), US, or one of 10 international countries.

The model is evaluated with **NDCG@5** — it may submit up to 5 ranked predictions per user, and is rewarded more when the correct answer appears higher in the ranking.

Why this matters to Airbnb

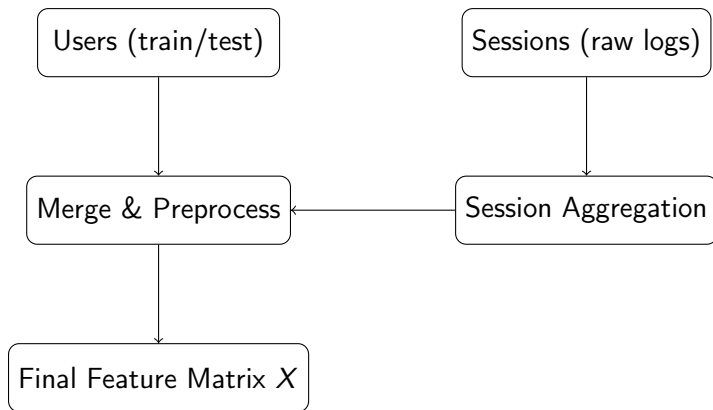
- Predict destination early → **personalise** the user experience
- Show relevant listings, currencies, travel tips before the user searches
- Higher conversion rates from first-session personalisation
- Originally posed as a **recruiting challenge** at Kaggle

Core difficulty

Over **87%** of users are either NDF (58%) or US (29%). The rarest class (PT) represents only **0.10%** of users — a 583:1 ratio.

Data Sources

- **Users table:** demographics + signup/account signals
- **Sessions table:** clickstream logs (actions, devices, times)
- Goal: aggregate sessions → user-level features and merge



Three input files

File	Size
train_users_2.csv	213,451 users
test_users.csv	62,096 users
sessions.csv	10.5M action rows
countries.csv	10 country profiles
Total unique users	275,547

Sessions table features (6)

user_id, action, action_type, action_detail, device_type, secs_elapsed

The sessions table must be **aggregated** from 10.5M rows into one row per user — 23 engineered features per user.

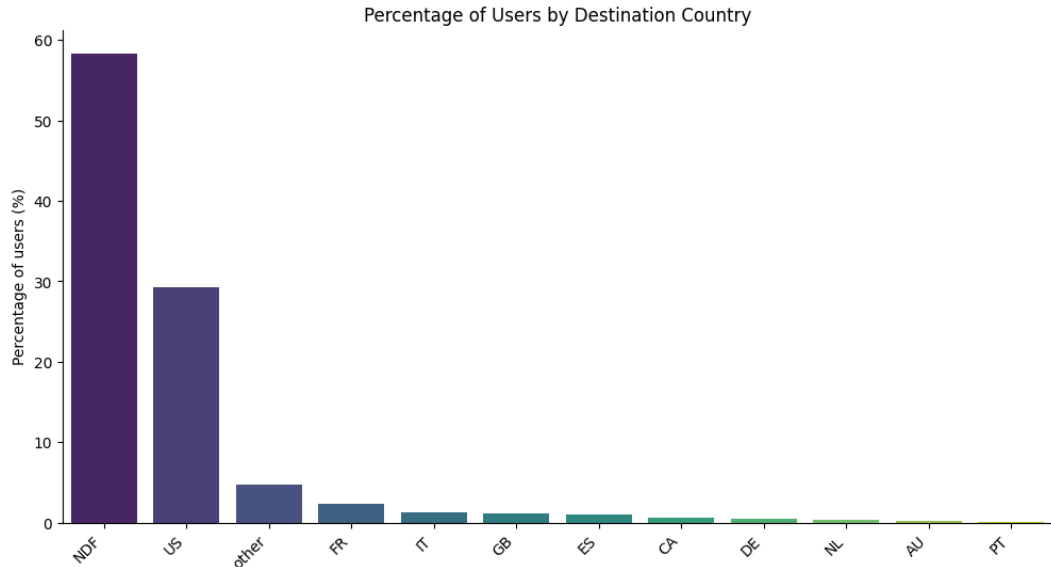
Users table features (16)

- **Demographics:** age, gender, language
- **Acquisition:** signup_method, affiliate_channel
- **Temporal:** date_account_created,

Key pipeline challenge

Only **49%** of users have any session records. The rest (primarily NDF users) must be retained with zero-filled session features via a **left join** — dropping them would introduce severe selection bias.

Target Distribution (Imbalance)



Target Distribution — Explanation

Class shares (training set, 213,451 users)

Destination	Share
NDF	58.35%
US	29.22%
other	4.73%
FR	2.35%
IT	1.33%
GB	1.09%
ES	1.05%
CA	0.67%
DE	0.50%
NL	0.36%
AU	0.25%
PT	0.10%

Imbalance severity

- **NDF + US = 87.6%** of all users
- Ratio of NDF to PT: **583:1**
- A naive model predicting NDF for all users gets 58% accuracy but $\text{NDCG@5} \approx 0.58$
- Most rare classes have fewer than **700 training examples**

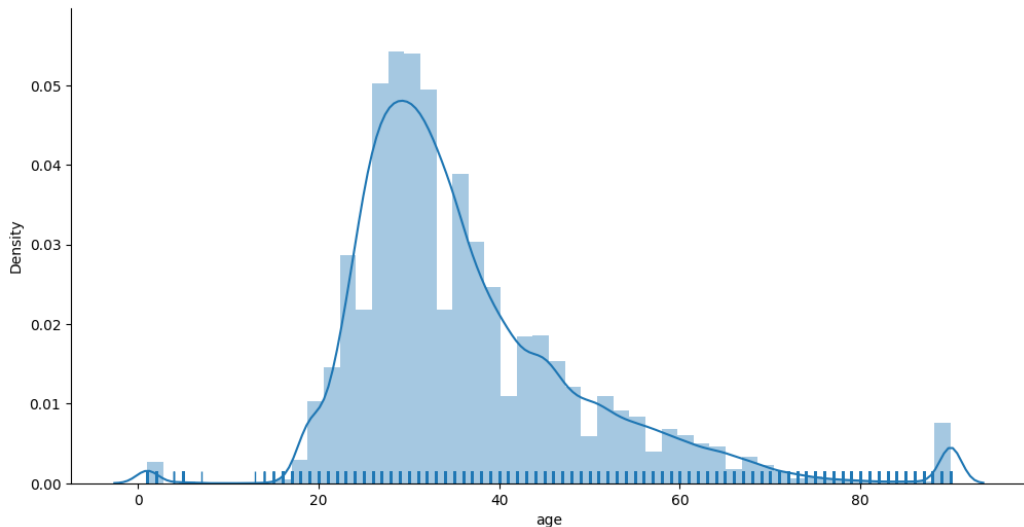
Consequences for modelling

- Standard class weighting **hurts NDCG@5** (proven experimentally — Chapter 9)
- Hierarchical decomposition solves this **structurally** instead

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Age Distribution



Age Distribution — Explanation

What the plot shows

The distribution of user ages across all 213,451 training users, after cleaning. Roughly **right-skewed** with a peak in the 25–40 range, consistent with Airbnb's core traveller demographic.

Age data quality issues

- **828 entries** stored as birth years — corrected
- Ages > 90 capped at 90
- **116,866 nulls** (45%) imputed with median age = 33
- age_group created **before** imputation

Why age matters

age_group is the **#1 feature** across all models:

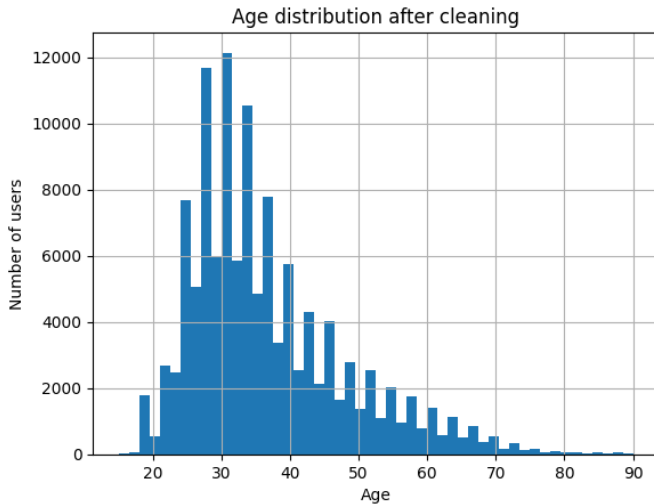
CatBoost gain	12.02%
XGB permutation	0.122
DT permutation	0.066
LR permutation	0.047

Younger users book international; older users skew toward NDF or US.

Feature engineering takeaway

Two representations used: continuous age and 5-bin ordinal age_group, both included in the final 132-feature matrix.

Age Histogram (Cleaned / Alternative View)



Age Histogram — Explanation

Cleaned vs. raw distribution

This alternative view shows the age distribution **after the 4-step cleaning pipeline**:

- 1 Detect and correct 828 year-format entries
- 2 Cap values > 90 at 90
- 3 Create `age_group` before imputation
- 4 Impute 116,866 nulls with median (33)

5-bin age groups

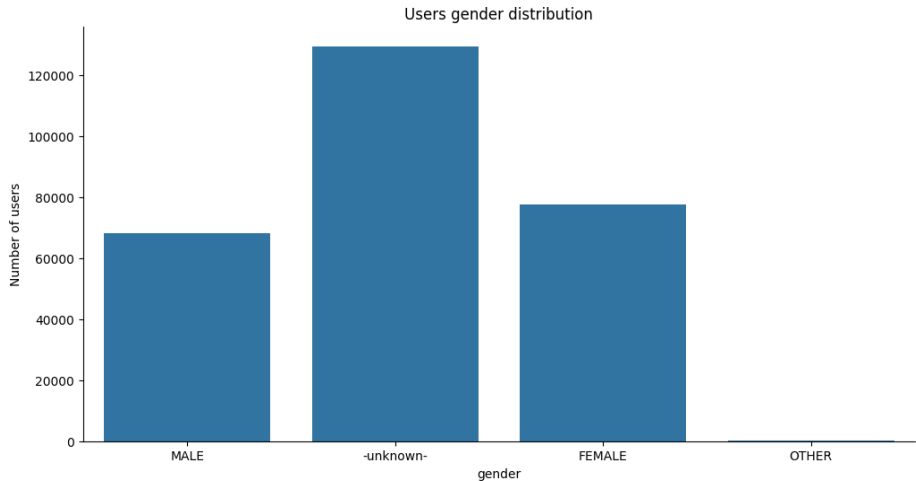
- Group 0** Age unknown (before imputation)
- Group 1** Age ≤ 25
- Group 2** Age 26–35
- Group 3** Age 36–50
- Group 4** Age > 50

Insight from the histogram

The cleaned histogram reveals a realistic bell-shaped core (25–45 years) with a long right tail. The “clumping” artifacts visible in the raw distribution (year-format entries all mapping to age 0 or 2014) are resolved after correction.

Preserving the **Group 0** (age was missing) bin is critical: unknown-age users have a systematically different booking rate from known-age users, making the missingness itself a predictive signal.

Gender Distribution



Gender Distribution — Explanation

Four gender categories

Value	Approx. share
Female	~25%
Male	~30%
Unknown	~45%
Other	<1%

Nearly half of all users have no gender on record. This is **not random** — unknown-gender users show a distinct behavioural pattern.

Missing gender signal

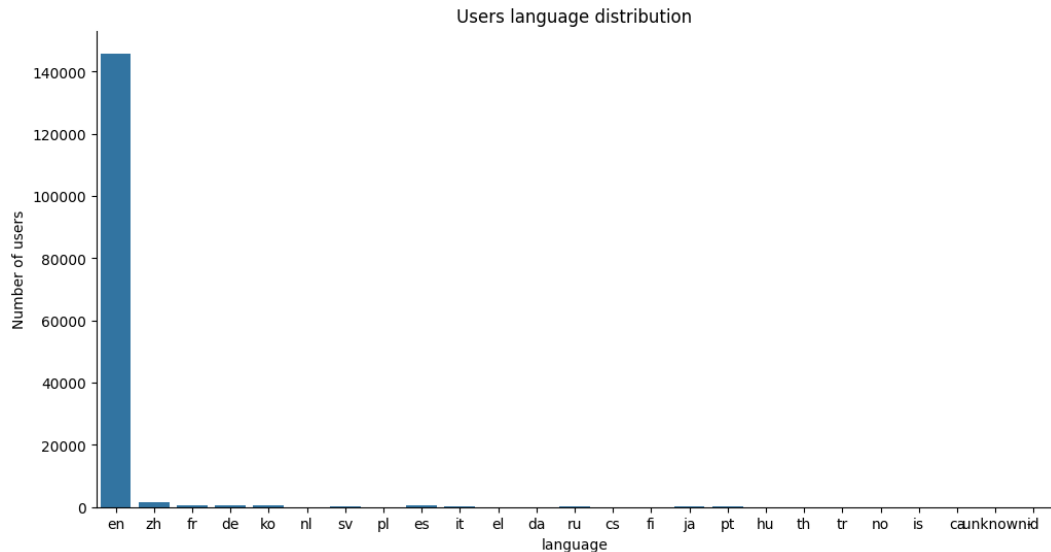
Unknown gender correlates strongly with NDF. Users who do not complete their profile are less engaged and less likely to book. The

Gender in modelling

Gender is a **moderate-to-high** importance feature depending on the model type:

- **Logistic Regression:** top-2 permutation features are gender_ -unknown- (0.054) and age_group (0.047)
- **Gradient boosting:** gender importance is lower but consistent in top-15
- **Stage C (international):** gender rises to top-10 — it helps discriminate *which* country among international travellers (European destinations show different gender distributions)

Language Distribution



Language Distribution — Explanation

What the plot shows

Distribution of the language feature across training users — the language they set in their Airbnb account.

97%+ of users have English set as their language. The next languages (French, Chinese, German, Spanish) appear in very small proportions.

Preprocessing decision

Languages with fewer than **2,000 users** are grouped into an "Other" bucket (2,000-count cutoff policy applied to all 13 categorical features).

Surprising finding

Despite an obvious theoretical relationship (a French-speaking user should be more likely to book FR), **language is the weakest predictor** in all models.

It appears in no model's top-20 gain importance and only marginally in permutation importance.

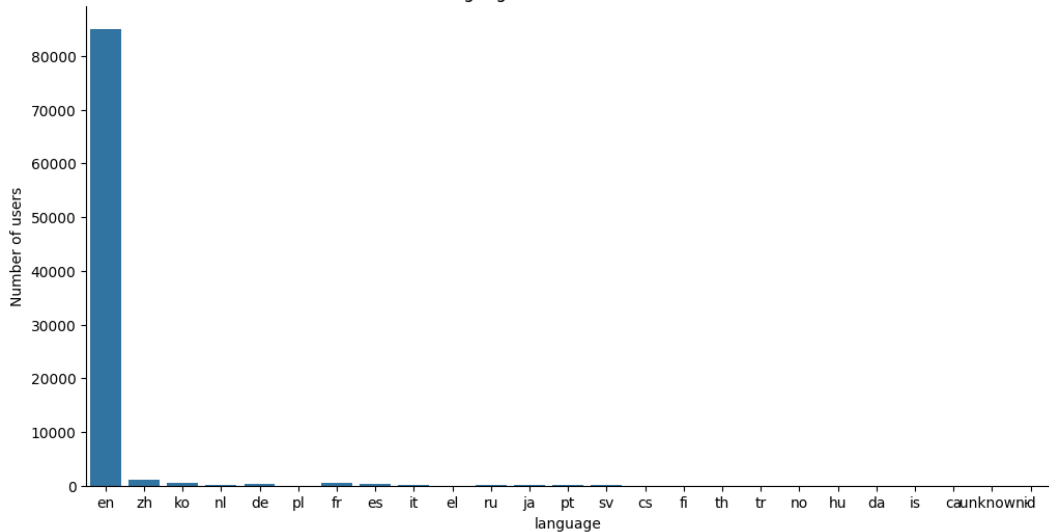
Reason: the 97% English concentration creates near-zero variance — the feature carries almost no distinguishing information.

Takeaway

High-cardinality categorical features with heavily skewed distributions add noise, not

Language Distribution (Excluding US)

Users language distribution without the US



Language Distribution (Excluding US) — Explanation

Why exclude US bookers?

The US class dominates at 29% of training users. By removing US bookings, this view exposes the language distribution among **international travellers only** — a much smaller and more linguistically diverse subset.

What this reveals

Among non-US bookers, non-English languages become more visible:

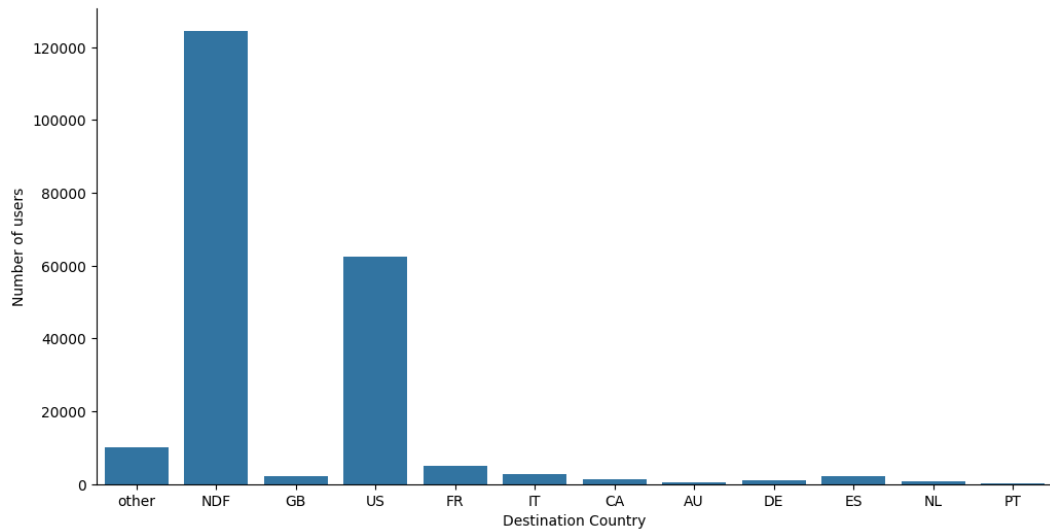
- French users skew toward FR
- German users skew toward DE
- Spanish speakers toward ES

Modelling implication

This EDA view informed the decision to **keep** language in the feature set (despite low importance) rather than dropping it — it may contribute marginal signal for the rarest international classes (PT, NL, AU).

The `not_English` binary flag derived from this appears at rank 20 in XGBoost gain importance — the only language-related feature to crack any top-20 ranking.

Destination Counts (Bookings)



Destination Counts — Explanation

What the plot shows

Raw count of training users per destination class. This directly visualises the class imbalance problem:

- NDF: $\sim 124,000$ users
- US: $\sim 62,000$ users
- other: $\sim 10,000$ users
- FR: $\sim 5,000$ users
- PT: only ~ 213 users

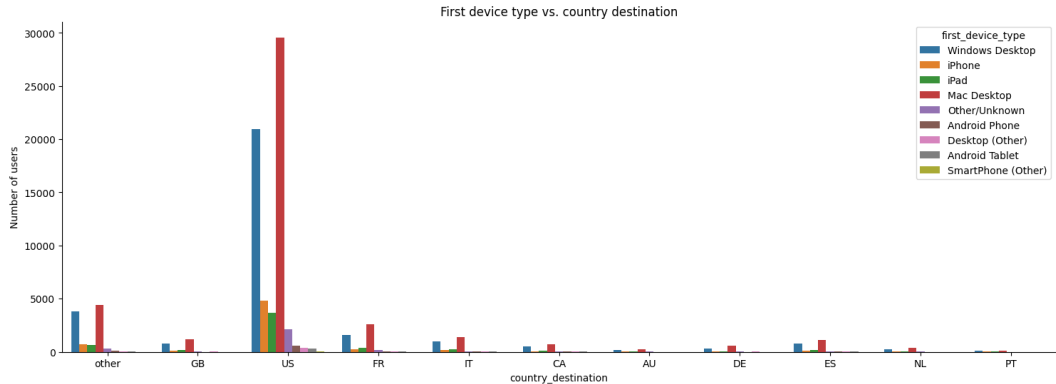
Key implication: gradient starvation

When training a flat 12-class model, rare classes like PT (213 examples) receive proportionally tiny loss gradients compared to NDF (124,000 examples).

The model learns to confidently predict the top-2 classes and nearly ignores the bottom-5.

Solution adopted: the 3-stage hierarchical architecture isolates international prediction into Stage C, where France grows from **2.35%** globally to $\approx$ **19.7%** of the training subset — without any synthetic oversampling.

Device Type vs Destination



Device Type vs Destination — Explanation

What the plot shows

Stacked or grouped bar chart showing the distribution of `first_device_type` across destination classes. Desktop browsers dominate; mobile and Mac/Windows splits also vary by destination.

Visible pattern

Mac users show a higher proportion of international bookings (particularly FR, IT, GB) relative to Windows users. Mobile users are over-represented in NDF.

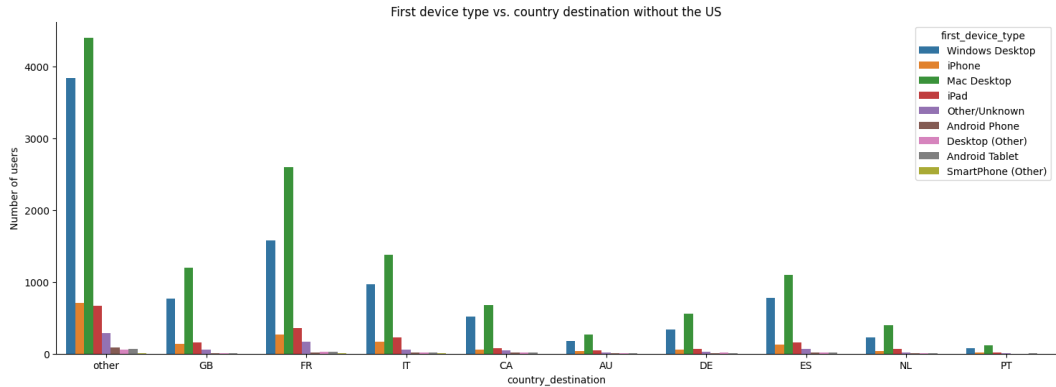
Feature engineering outcome

Device type is encoded both as:

- **One-hot dummies** from `first_device_type` (user table)
- **4 binary device category flags** from sessions (Apple / Desktop / Mobile / Tablet) derived from `session device_type`

Both sets are included in the 132-column matrix. Device features rank as **Low–Moderate** in importance — useful but far below age and temporal signals.

Device Type vs Destination (Excluding US)



Device Type vs Destination (Excl. US) — Explanation

Why the US-excluded view matters

With US removed, the device–destination relationship among international travellers becomes clearer. International bookers show more device diversity and a higher proportion of Mac/iPhone users, consistent with higher-income travellers.

Specific findings

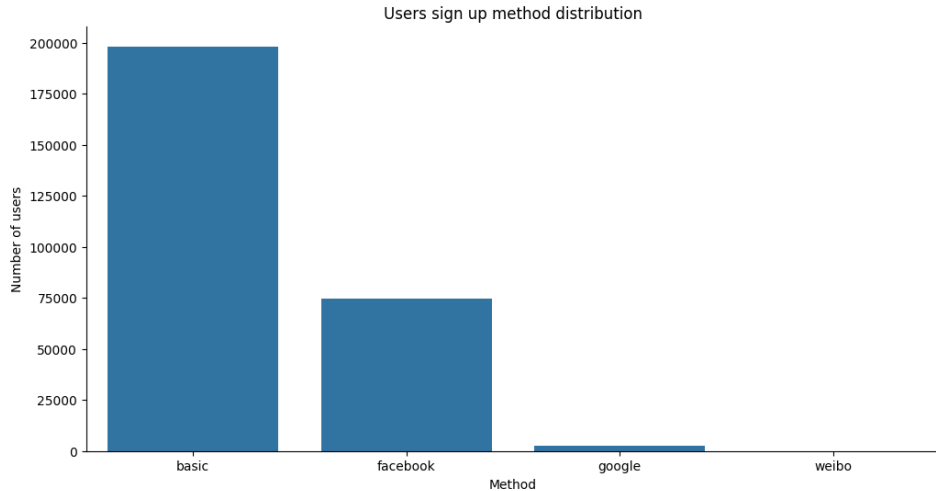
- FR, IT bookers: higher Mac proportion
- other and rare destinations: more mixed device profiles
- iPhone users more likely to be international

Stage C relevance

This pattern is most relevant to **Stage C** of the hierarchical model — predicting *which* international country for confirmed international travellers.

The `first_device_type` dummies do appear in Stage C's top-20 feature importances, though at lower ranks than age and temporal features.

Signup Method Distribution



Signup Method Distribution — Explanation

Three signup methods

Method	Approx. share
basic (email)	~70%
facebook	~25%
google	~5%

Facebook signup was heavily promoted during Airbnb's growth phase (2011–2013).

Model importance

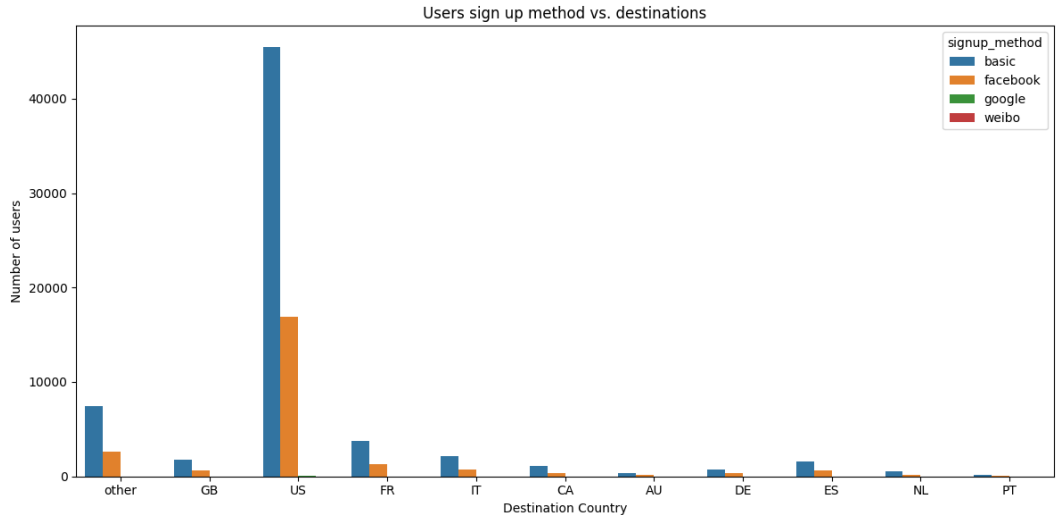
signup_method_facebook is a **top-10 feature** in multiple models:

- CatBoost Optuna gain: rank 6 (3.28%)
- XGBoost gain: rank 1 (0.065)

Why signup method is predictive

- 1 **Demographics proxy:** Facebook users in 2010–2014 skewed younger and more internationally mobile
- 2 **Engagement signal:** connecting a social account indicates higher platform investment
- 3 **Profile completeness:** Facebook signup auto-fills more profile fields (age, gender) — leading to fewer NDF outcomes

Signup Method vs Destination



Signup Method vs Destination — Explanation

What the plot shows

Conditional distribution of destinations for each signup method. Facebook users book at a **significantly higher rate** and show a higher share of international destinations compared to basic email signups.

Key observation

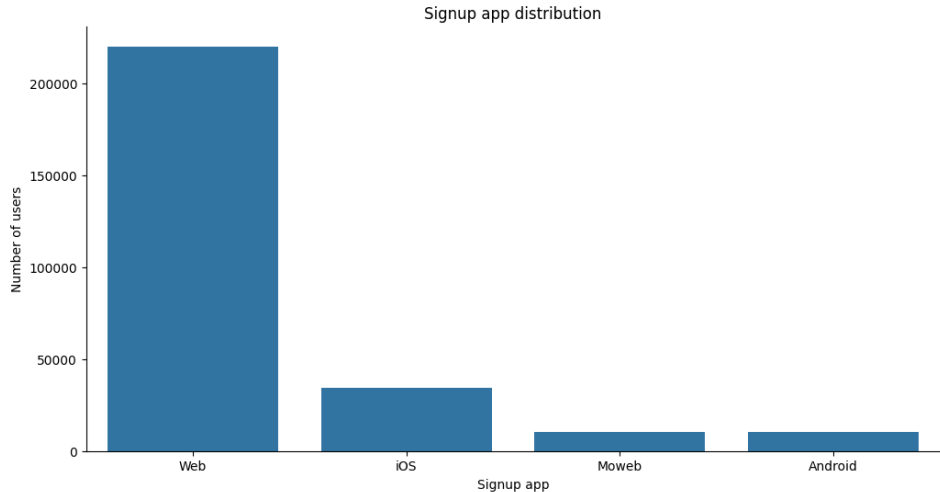
- basic signups: highest NDF share — users who sign up with email alone are more likely to browse without booking
- facebook signups: lower NDF, higher US and international share

Interaction effect

The signup method interacts with age: younger Facebook users show the highest international booking rate of any group.

This interaction is naturally captured by gradient-boosted trees (which discover feature interactions implicitly through tree splits), but is **missed entirely** by Logistic Regression — contributing to the large gap between these two model classes.

Signup App Distribution



Signup App Distribution — Explanation

Four signup applications

App	Approx. share
Web	~80%
iOS	~12%
Android	~5%
Moweb	~3%

Web is dominant (most users signed up via desktop browser in this 2010–2014 dataset).

Feature behaviour

signup_app_Web is the **only signup-app dummy to appear** in a top-15 permutation importance ranking — uniquely in the Decision Tree (rank 2, importance 0.041).

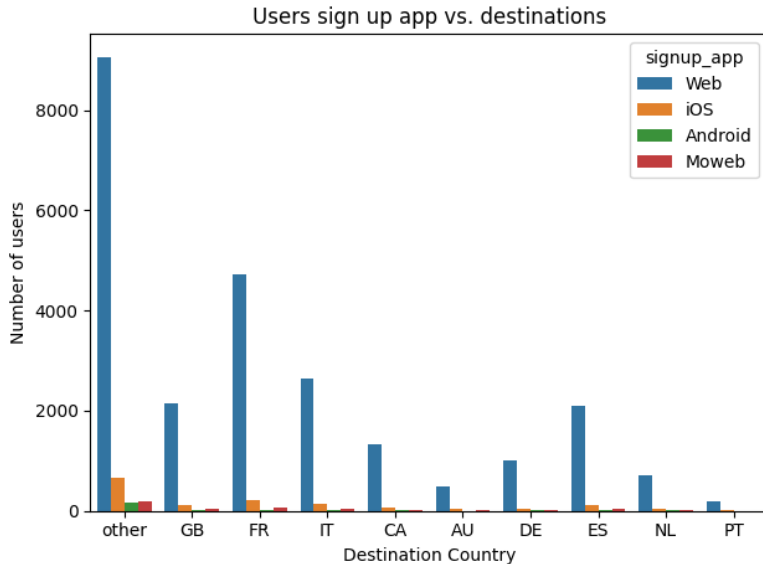
Mobile vs. web signing up

Mobile app signups (iOS/Android) correlate differently with destination than web signups:

- iOS users show marginally higher international booking rates
- Android users closer to web users in booking patterns
- Moweb (mobile web browser) users behave similarly to NDF

The signup app feature interacts with the session device type features, but the individual signal is modest.

Signup App vs Destination



Signup App vs Destination — Explanation

Conditional destination distribution

This plot shows destination shares conditional on each signup app.

The most striking pattern: **iOS users have the lowest NDF rate** of any app group — suggesting that mobile-first users who do sign up are more committed to completing a booking.

Android vs. iOS comparison

Android users have a higher NDF rate than iOS users in this dataset, consistent with Airbnb's 2010–2014 userbase where iOS early adopters were more affluent and internationally mobile.

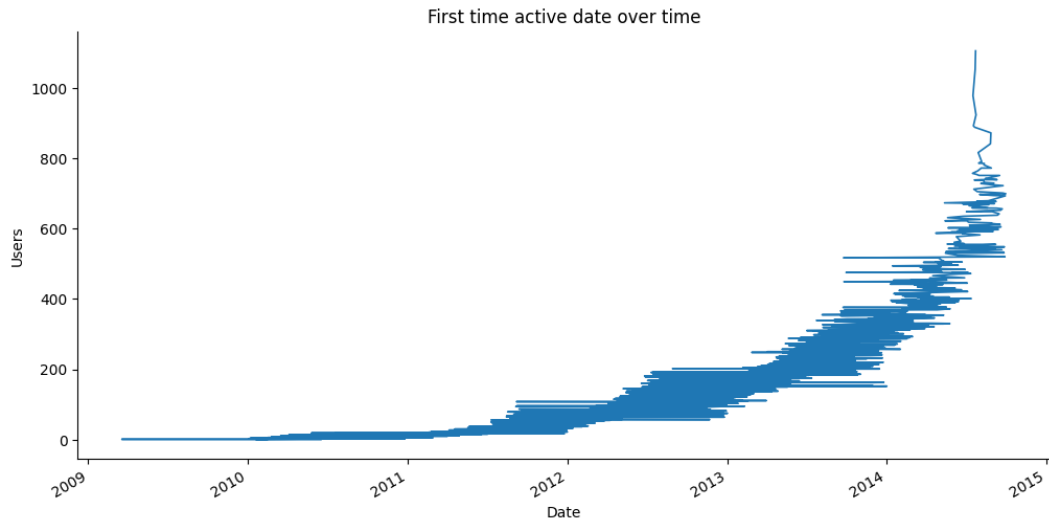
Context: 2010–2014 data

This dataset captures a specific historical moment when:

- iPhone 4/5 era — iOS users were a premium demographic
- Android was less prevalent among international travellers
- Web was the dominant booking channel

These device–destination patterns may not generalise to modern data, which is one reason the report flags temporal snapshot as a **key limitation** in Chapter 15.

Temporal Signals



New accounts per month

Temporal Signals — Explanation

Two temporal series shown

- **First active over time:** timestamp of first website activity — the earliest signal of a user's existence
- **Account created over time:** when the user formally registered

Both show **exponential growth** from 2010 to 2014 — Airbnb's hyper-growth phase.

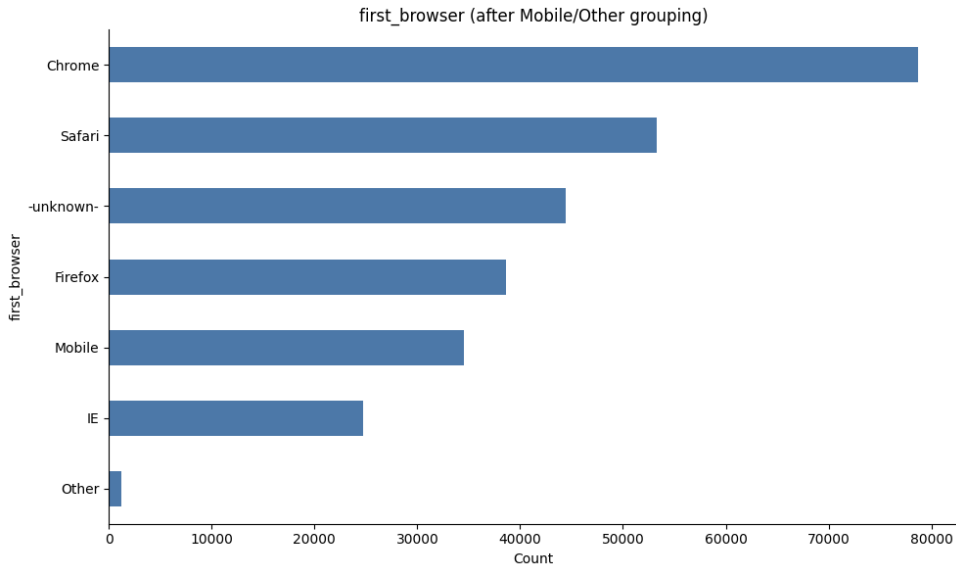
10 temporal features derived

Year, month, weekday decomposition of each date, plus US holiday flag and business day flag. These 10 features are among the **top-10 most important** in all gradient-boosted models.

Why temporal features predict destinations

- **Year:** Platform maturity — later cohorts (2013–2014) book internationally at higher rates
- **Month:** Travel seasonality — summer signups lead to more bookings; winter signups more NDF
- **Weekday:** Engagement pattern — weekend signups correlate with impulse browsing and NDF
- **month_first_active** ranks **3rd** in CatBoost gain (4.36%) and 2nd in LightGBM split count.

First Browser Distribution



First Browser Distribution — Explanation

What the plot shows

Distribution of `first_browser` — the browser used during the user's first recorded session.

Chrome and Safari dominate. Firefox, IE, and Mobile Safari appear in smaller proportions. A long tail of rare browsers is grouped into "Other" by the 2,000-count cutoff.

Preprocessing applied

Mobile browser variants (mobile Safari, mobile Chrome) were consolidated with their desktop counterparts before one-hot encoding to reduce dimensionality and improve generalisation.

Browser as an indirect signal

Browser choice reflects device type and indirectly reflects:

- **OS:** Safari → Mac/iPhone; Chrome/Firefox → Windows/Android
- **Technical sophistication:** Firefox users show higher engagement
- **Era:** IE usage peaks in 2010–2012; Chrome rises sharply 2013–2014

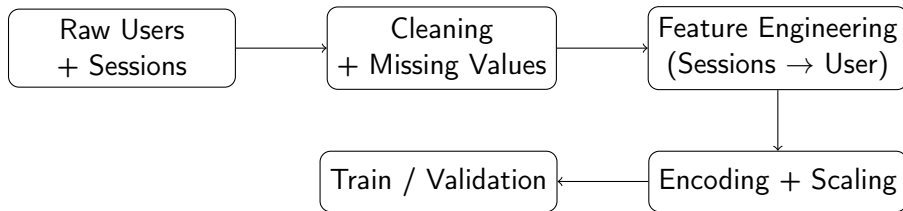
First browser has **Low–Moderate** importance in gradient-boosted models — mainly useful as a proxy for device type, which is already captured by `first_device_type`.

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Preprocessing Pipeline

- Handle missing values (categorical + numerical)
- Age cleaning / outlier filtering
- One-hot / encoding for categorical attributes
- Session log aggregation to user-level statistics



Preprocessing Pipeline — Explanation

Sessions cleaning (10.5M rows)

- Drop **34,496** rows — null `user_id`
- Fill **79,480** null `action` → "missing"
- **Two-pass** impute `action_type` & `action_detail`:
mode within action groups → global mode fallback
- `secs_elapsed` nulls → median per user

Users cleaning

- Categorical nulls → "missing"
- Continuous nulls → 0
- 13 features: 2,000-count cutoff for rare category grouping

Final feature matrix

Property	Value
Total rows	275,547
Features	132
Null values	0
Train (80%)	170,760
Val (stratified, 20%)	42,691
One-hot encoded cols	~100
Duration stats	9
Unique-count cols	4
Device flag cols	4
Temporal cols	10
Age cols	2

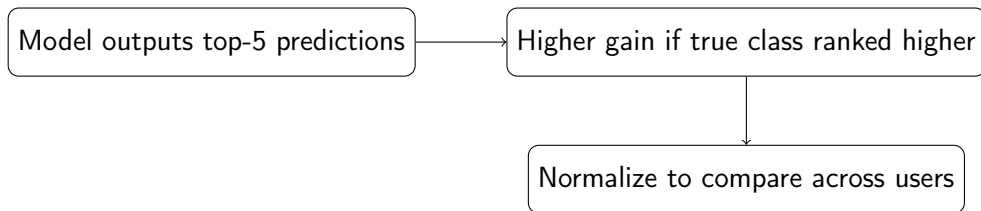
Left join preserves 140,064 sessionless users.
Dropping them would create selection bias toward

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Evaluation Metric: NDCG@5

- Competition metric: **NDCG@5** (rank-based, rewards correct class in top-5)
- Appropriate when producing **probability rankings** of destinations
- Complemented with: Accuracy, Macro-F1, Weighted-F1 (as in report)



NDCG@5 — Formula & Example (1/2)

Formula

$$\text{NDCG@5} = \frac{1}{n} \sum_{i=1}^n \frac{\text{DCG@5}_i}{\text{IDCG@5}_i}$$

$$\text{DCG@5} = \sum_{k=1}^5 \frac{2^{r_k} - 1}{\log_2(k + 1)}$$

$r_k = 1$ if the true destination is at rank k , else 0.

A perfect model (rank-1 always correct) scores **1.0**.

Rank-1 correct gives **2.58× more credit** than rank-5 correct — the model must be confident.

Example: true destination = FR

Rank	Predicted	DCG
1	FR	1.000 (full)
2	FR	0.630
3	FR	0.500
4	FR	0.431
5	FR	0.387
—	not in top-5	0.000

Why this matters here

NDF + US = **87.6%** of users. NDCG@5 heavily rewards correct rank-1 predictions on the dominant classes — penalising any model that reduces confidence on them.

NDCG@5 — Metrics Suite & Tensions (2/2)

6 metrics tracked for every model

Metric	Focus
NDCG@5	Primary: ranking quality
Accuracy	Top-1 correctness
Macro F1	Equal weight per class
Weighted F1	Frequency-weighted F1
Macro Recall	Per-class recall (fairness)
Top-5 Accuracy	Any hit in top-5

Fundamental tension

Improving Macro F1/Recall requires boosting rare class confidence — which **directly hurts** NDCG@5 by reducing confidence on NDF/US.

These two groups of metrics **cannot both be optimised simultaneously** with resampling.

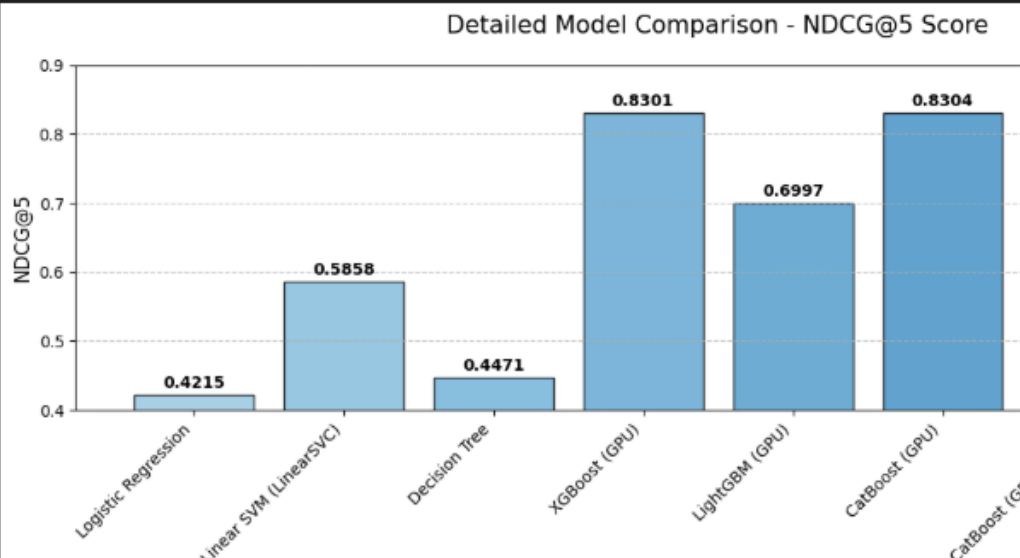
Confirmed by our experiments

Strategy	NDCG@5	Mac.F1
CatBoost default	0.8304	0.107
Hybrid SMOTE	0.8269	0.109
Balanced Bootstrap	0.8041	0.114
Hierarchical	0.8610	0.120

The solution

Both SMOTE strategies improve Macro F1 but degrade NDCG@5. The **hierarchical decomposition** is the only approach that improves all 6 metrics simultaneously — by solving the imbalance **structurally** rather than statistically.

Model Comparison (NDCG@5)



Model Comparison — Explanation

Three performance tiers

Tier	Models	NDCG@5
1	LR, DT, SVM	0.42–0.59
2	LightGBM, XGB, CatBoost	0.70–0.83
3	Hierarchical, Ensemble	0.86–0.87

Leaderboard (top 6)

2-Model Ensemble	0.8651
Hierarchical CB	0.8610
3-Model Ensemble	0.8595
CatBoost Optuna	0.8307
CatBoost default	0.8304
XGBoost	0.8301

Key insight: tier gaps

- Tier 1 → 2 gap: **+0.11**
(SVM to LightGBM — gradient boosting is non-negotiable)
- Tier 2 → 3 gap: **+0.03**
(flat CatBoost to Hierarchical — structural decomposition)
- Within Tier 3: **+0.004**
(ensemble increment, confirmed by CV)

Main lesson

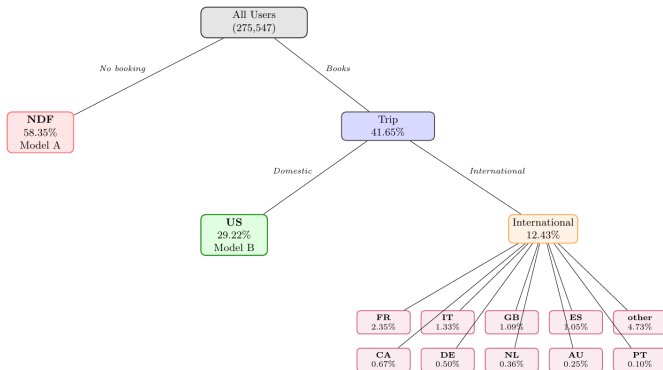
Hyperparameter tuning (Optuna, 20 trials) added only **+0.0003**. The hierarchical structure added **+0.0306** — **100× more**.

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Hierarchical Target Structure

- Level 1: NDF vs Booking
- Level 2: US vs International
- Level 3: International destination (10-class)



Hierarchical Target Structure — Explanation

3-stage CatBoost pipeline

Stage	Task	Loss	Iter.
A	Trip vs NDF	Logloss	600
B	US vs Intl	Logloss	600
C	10-class	MultiClass	800

Probability combination

Chain rule applied:

$$P(\text{NDF}) = 1 - P(\text{Trip})$$

$$P(\text{US}) = P(\text{Trip}) \cdot P(\text{US} \mid \text{Trip})$$

$$P(c_k) = P(\text{Trip}) \cdot P(\text{Intl} \mid \text{Trip}) \cdot P(c_k \mid \text{Intl})$$

All 12 probabilities sum to 1 by construction.

Why decomposition beats flat training

- Stage A: only needs to learn one clean binary decision — no dilution from 11 booking classes
- Stage C: France grows from 2.35% globally to **≈19.7%** of Stage C's subset — rare classes get proportional gradient without SMOTE
- Each specialist model **optimises a different loss** appropriate to its sub-problem

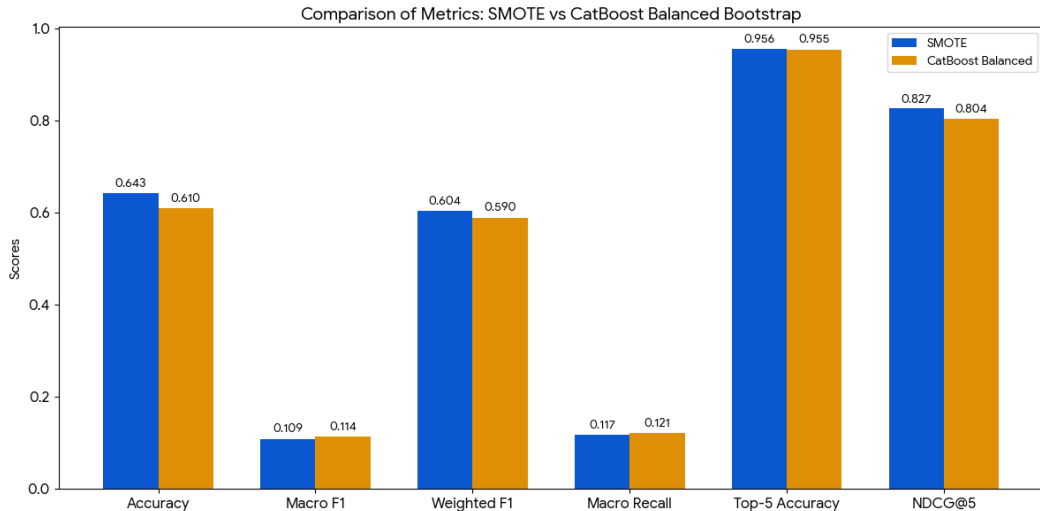
Result

$$\text{NDCG@5} = 0.8610$$

+0.0306 over best flat model (0.8304)

Improves **all 6 metrics** simultaneously

Imbalance Handling Experiments



Imbalance Handling — Explanation

Two strategies tested

Hybrid Sampling (SMOTE + RUS):

- Phase 1: RandomUnderSampler → NDF: 35%, US: 25%
- Phase 2: SMOTE ($k = 5$) → minorities → 4% each
- PT grows 0.10% → 3.97% (40×)
- Final: 172,000 rows

Balanced Bootstrap:

- 15,000 rows per class with replacement
- Perfect 1:1 ratio, 51,225 rows total
- PT bootstrapped 87× from 172 real examples

Strategy	NDCG@5	Δ
CatBoost baseline	0.8304	—
Hybrid Sampling	0.8269	−0.003
Balanced Bootstrap	0.8041	−0.026
Hierarchical	0.8610	+0.031

Why resampling hurts NDCG@5

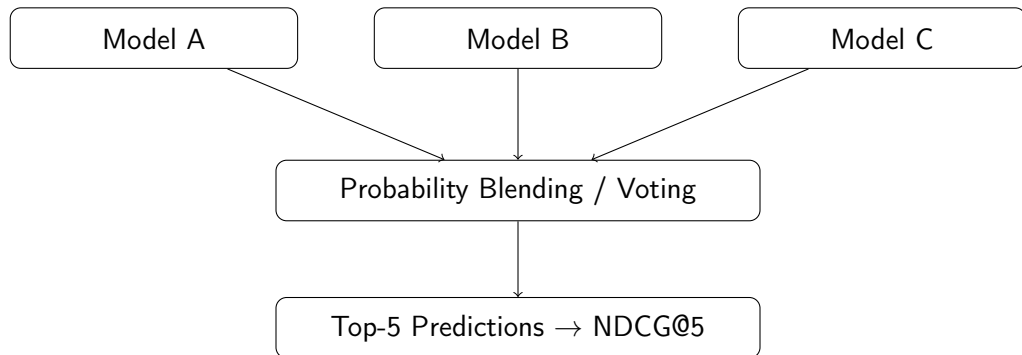
NDCG@5 rewards confident correct rank-1 predictions. NDF + US = 87.6% of validation. Reducing model confidence on these classes (by forcing balanced training) degrades ranking for the **majority of users**. Hierarchical decomposition solves the structural problem **without this trade-off**.

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Ensemble Strategy (High Level)

- Combine strong learners (e.g., CatBoost + XGBoost + Hierarchical model)
- Blend probabilities to improve top-5 ranking stability
- Optuna used for hyperparameter optimisation (as in report)



Ensemble Strategy — Explanation

Two ensemble configurations tested

2-Model Ensemble (selected):

$$p = \frac{0.70 \cdot p_{\text{Hier}} + 0.30 \cdot p_{\text{CatOpt}}}{2}$$

- Hierarchical provides structured conditional probability chain
- CatBoost Optuna provides joint softmax over 12 classes
- Diverse error modes → variance reduction

3-Model (tested, worse):

$$0.60 \cdot p_{\text{Hier}} + 0.25 \cdot p_{\text{XGB}} + 0.15 \cdot p_{\text{Cat}}$$

Adding XGBoost dilutes stronger models.

Optuna hyperparameter tuning

20 trials over CatBoost search space: iterations (400–1200), learning_rate (0.02–0.15), depth (6–10), 12_leaf_reg (1–20), random_strength, bagging_temperature.

Best: iter=705, lr=0.0864, depth=6.

Gain: +**0.0003** NDCG@5 over default.

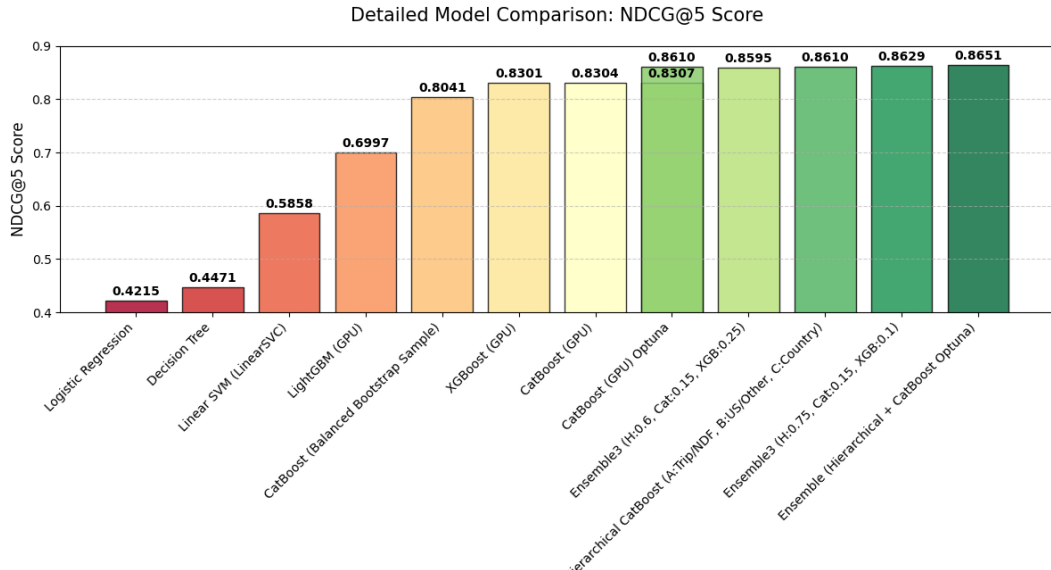
Cross-validation stability

5-fold stratified K-Fold on ensemble pipeline:

Mean = **0.8305**, Std = **0.0002**, CV% = **0.027%**

Confirms no overfitting to validation split.

Detailed Model Comparison



Detailed Model Comparison — Explanation

All 12 configurations compared

Model	NDCG@5
2-Model Ensemble	0.8651
Hierarchical CatBoost	0.8610
3-Model Ensemble	0.8595
CatBoost (Optuna)	0.8307
CatBoost (default)	0.8304
XGBoost	0.8301
CB Hybrid Sampling	0.8269
CB Balanced Bootstrap	0.8041
LightGBM	0.6997
Linear SVM	0.5858
Decision Tree	0.4471
Logistic Regression	0.4215

NDCG@5 progression (+0.4436 total)

Logistic Reg. (baseline)	0.4215
→ SVM	+0.1643
→ Gradient Boosting	+0.2446
→ Optuna Tuning	+0.0003
→ Hierarchical	+0.0303
→ Ensemble	+0.0041

Final	0.8651
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Share of total gain

Gradient boosting: **55%** of total gain
Linear upgrade (LR→SVM): **37%**
Hierarchical structure: **7%**
Tuning + ensemble: **<1%**

Final model: Hierarchical CatBoost + CatBoost Optuna (0.70/0.30)

Final Model Performance — Explanation

Final model configuration

Property	Value
Constituent 1	Hierarchical CatBoost
Constituent 2	CatBoost (Optuna)
Weights	0.70 / 0.30
Training data	213,451 users (full)
Feature matrix	$213,451 \times 132$
CV NDCG@5	0.8305 ± 0.0002

Full validation metrics

Accuracy	0.7422
Macro F1	0.1233
Weighted F1	0.6817
Macro Recall	0.1291
Top-5 Accuracy	0.9586

Total journey

Logistic Regression baseline
0.4215
↓ **+0.4436**
Final 2-Model Ensemble
0.8651

Why the 2-model outperforms 3-model

The 3-model ensemble (NDCG@5 = 0.8595) adds XGBoost (0.8301) which is weaker than both constituents. The extra model dilutes the hierarchical model's strong structural signal, reducing the ensemble below the 2-model result.

Agenda

- 1 Problem & Data
- 2 Exploratory Data Analysis
- 3 Preprocessing & Feature Engineering
- 4 Modeling & Evaluation
- 5 Hierarchical Approach & Imbalance Handling
- 6 Ensembles & Final Results
- 7 Conclusions

Key Takeaways

- Session aggregation provides strong behavioral signals beyond demographics
- Boosting models (CatBoost / LightGBM / XGBoost) outperform simple baselines
- Hierarchical decomposition aligns with the semantic structure of the target
- Imbalance handling and ensembling improve top-5 ranking quality (NDCG@5)

Key Takeaways — In Depth

1 Architecture $\gg$ Hyperparameters.

Hierarchical decomposition: +0.0306. Optuna tuning (20 trials): +0.0003. Understanding the problem structure is $100\times$ more impactful than searching over parameter space.

2 Gradient boosting is non-negotiable.

No linear or shallow model exceeded $\text{NDCG@5} = 0.59$. Non-linear interactions (age $\times$ temporal $\times$ session) can only be captured by tree ensembles.

3 Resampling hurts NDCG@5.

Both SMOTE and balanced bootstrap degraded ranking. The hierarchical model improved all 6 metrics simultaneously — a strictly better solution.

4 Age dominates all models.

age_group ranks #1 across every model and every importance method. Combined with temporal registration features and session duration, three feature groups carry the majority of signal.

5 Cross-validation confirms stability.

CV std = 0.0002 over 5 folds — the final score of 0.8651 is a genuine, reproducible result.

- Richer sequential modeling of sessions (time-aware features / embeddings)
- Calibration of predicted probabilities for better ranking stability
- More systematic feature selection / interaction discovery

Feature Engineering

- **Richer age representation:** decade bins, spline transforms, $\text{age} \times \text{signup_method}$ interactions
- **Sequence-aware sessions:** RNN or attention over ordered action sequences — captures booking intent path for rare international destinations
- **Temporal cohort features:** $\text{month} \times \text{year}$ interaction, business-day vs. holiday flags

Modelling

- **Calibrated probabilities:** Platt scaling / isotonic regression — necessary for any deployment that interprets probabilities directly
- **Stage C specialist models:** few-shot or prototypical networks for the rarest destinations (PT 0.10%, AU 0.25%, NL 0.36%)
- **Stacking / meta-learning:** train a meta-learner on out-of-fold predictions instead of fixed weights
- **Temporal split validation:** train on 2010–2013, test on 2014 to simulate deployment conditions

Thank you!

Questions?

Airbnb New User Bookings

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Final NDCG@5 = **0.8651** (+0.4436 over baseline)

github.com/Rezacs/Airbnb